AKASH REDDY

SENIOR FULL STACK DEVELOPER

C#/.NET • React 18 • WPF/MVVM • Industrial Automation

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PROFESSIONAL SUMMARY

Senior Software Engineer with 4 years 10 months of experience building enterprise and industrial automation software using C#, .NET, React and Windows desktop technologies. Hands-on across end-to-end flows from React/WPF user interfaces to ASP.NET Core REST APIs, real-time SignalR updates, EF Core/PostgreSQL data access and TCP/IP-based equipment integration. Strong in async programming, dependency injection, SOLID design, production debugging, performance-focused development, NUnit testing and Agile delivery. Practical Azure DevOps/YAML CI/CD exposure and active use of AI-assisted engineering for debugging, test design and documentation.

CORE TECHNICAL SKILLS

Backend & C# C#, .NET 8, .NET Framework, ASP.NET Core Web API, REST APIs, LINQ, generics, delegates/events, async/await, Task/TPL, middleware, dependency injection	**Frontend & Desktop** React 18, Redux Toolkit, hooks (useState/useEffect), JavaScript, HTML5/CSS3, Axios, WPF, XAML, MVVM, data binding, ICommand, INotifyPropertyChanged, ObservableCollection, WinForms
Architecture & Integration SignalR, TCP/IP equipment integration, client/server communication, event-driven workflows, OOP, SOLID, design patterns, scalable API and microservices concepts	**Data** Entity Framework Core, PostgreSQL, SQL, SQL Server/T-SQL fundamentals, query optimization, transactions and concurrency concepts
Testing & DevOps NUnit, unit/integration testing, code coverage, Git, Azure DevOps, YAML CI/CD, Visual Studio, Swagger, Postman, NLog	**AI-assisted Engineering** ChatGPT for debugging, test scenarios and documentation; familiarity with AI agents, MCP, A2A, context engineering and GitHub Copilot/Claude workflows

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Qusonic Technologies

Oct 2024 - Present

Halo Line Automation (ALI) | Industrial automation platform | C# | .NET 8 | React 18 | SignalR | EF Core | PostgreSQL

- Develop and maintain ASP.NET Core REST services for equipment commands, operator actions, jobs, alarms, events, interlocks and material tracking; integrate EF Core/PostgreSQL and keep controllers focused on HTTP concerns while services handle business logic.
- Built React/Redux equipment dashboards and operator-control workflows, integrating REST APIs and SignalR to keep machine state, production data and operator actions synchronized in real time.
- Implemented backend SignalR broadcasts and cached-state delivery for newly connected clients, improving consistency between equipment-status services and operator interfaces.
- Supported Online/Offline equipment control and TCP/IP / Beckhoff-based Ethernet communication; traced asynchronous connection-state transitions, delays and state-consistency issues across UI, API, services and equipment layers.
- Delivered load-port door request/cancel/confirm workflows, robot-station validation, equipment-control states, interlocks and history/tracking behavior driven by backend events.
- Investigate performance and production defects using logs and end-to-end tracing; apply dependency injection, async programming, error handling and separation of concerns; collaborate with frontend, backend and QA teams through reviews and Agile delivery.

EARLIER PROFESSIONAL EXPERIENCE

Software Engineer | ALTEN GT

Oct 2021 - Jul 2024

Progressed from Engineer Trainee - Software to Software Engineer; worked on Windows engineering, navigation-device and aerospace test applications.

SMARTNAV Manager - Windows Navigation Application | C# / WPF / XAML / .NET

- Developed WPF/XAML desktop features for geospatial map visualization using shapefiles, raster images and Canvas rendering, including zoom, navigation, route creation and waypoint management.
- Worked with data binding, command/event workflows, validation and view-logic separation aligned with MVVM; handled long-running device/file-transfer workflows with progress and error handling.
- Implemented creation, validation and transfer of routes, waypoints and map files to Android-based SmartNav devices; investigated UI/integration defects and supported responsiveness, testing and maintenance releases.

777 Audio System Simulator - Aerospace Test Stand | C# / WinForms / .NET Framework

- Maintained a Windows engineering simulator used in an aerospace test environment; delivered UI/workflow enhancements and investigated crashes, resource-handling defects and application-stability issues.
- Collaborated with QA on verification and maintenance releases, using NUnit/code coverage practices and structured root-cause analysis.

ARM Functional Safety Library | C++ / Linux / ISO 26262

- Contributed implementation, unit tests, qualification documentation and peer reviews in a traceable engineering environment with coding and safety standards.

SELECTED ENGINEERING & DELIVERY EXPOSURE

- Azure / CI-CD: Personal hands-on practice deploying an ASP.NET Core application and creating Azure DevOps YAML pipelines for restore, build, test, publish and deployment workflows.
- Architecture & reliability: REST/SignalR client-server design, async I/O, cancellation/timeouts concepts, race-condition analysis, scalable/stateless API fundamentals, SQL query optimization and distributed-system resilience concepts.
- AI-assisted development: Use ChatGPT for debugging approaches, code understanding, test-case ideas and documentation; currently strengthening agentic workflows, MCP, A2A and context-engineering concepts for SDLC automation.

SELECTED TECHNICAL HIGHLIGHTS

End-to-end full stack React/WPF UI -> ASP.NET Core APIs -> services -> EF Core/SQL -> equipment or external integrations.	**Desktop & responsiveness** WPF/XAML, MVVM concepts, binding/commands, async workflows, UI responsiveness and resource-defect investigation.
Industrial networking TCP/IP-based equipment communication, connection-state handling, real-time status/event flows and client/server troubleshooting.	**Cloud & AI readiness** Azure DevOps YAML CI/CD practice plus AI-assisted debugging/testing/documentation and MCP/A2A/context-engineering concepts.

EDUCATION

Bachelor of Engineering - Electronics and Communication Engineering

Aug 2021 | CGPA 8.26

Dr. Ambedkar Institute of Technology, Bengaluru